

Dynamic Charging Of Unmanned Aerial Vehicles Using IRS

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Abstract: This paper gives an insight into the feasibility of wireless charging of UAV using IRS and study its optimum conditions in which the system can be developed practically.

1. INTRODUCTION

❖ Wireless charging of UAV to increase durability and range of operation.

❖ **Challenges :-**

- UAV is generally battery operated and a mobile device.
- Limited duration of flight and energy constrained.

❖ **Objective :-**

- Obtain optimum conditions by which we can wirelessly harvest energy.
- Use of IRS to increase range beyond LoS and decrease outage probability.

❖ **Our Contribution :-**

- Derive CDF of outage probability w.r.t threshold energy and study of outage probability according to change in number of IRS elements.
- Change in outage probability with change in elevation angle of UAV w.r.t BS.

❖ **Related literature :-**

- Channel loss modelling and working of IRS is studied in [2].
- Wireless energy transfer using MIMO and its feasibility is provided in [1] with environments parameters in [3].

2. MOTIVATION AND PROBLEM FORMULATION

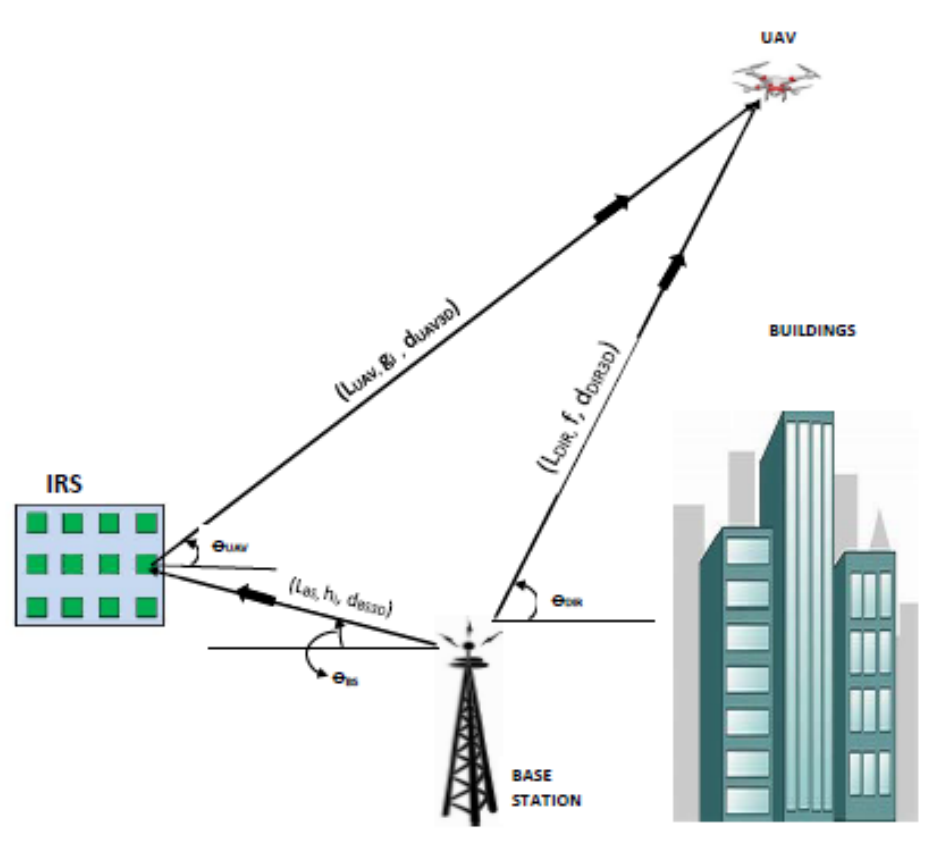


Figure 1: System Model

❖ Fig. 1 illustrates the system model by which UAV can be charged wirelessly where IRS can provide alternate path for energy transfer and increase its range of working.

- **Small scale fading :-** $h_i \sim CN(0,1)$ $g_i \sim CN(0,1)$ $f \sim CN(0,1)$
- **In terms of envelopes and phases :-** $h_i = \beta_i e^{-j\phi_i}$ $g_i = \alpha_i e^{-j\omega_i}$ $f = \gamma e^{-j\theta}$

• **Signal received BS $\rightarrow$ IRS $\rightarrow$ UAV:-** $r1 = \left[\sum_{i=1}^N h_i \sqrt{L_{BS,i}^{-1}} e^{j\phi_i} g_i \sqrt{L_{UAV,i}^{-1}} \right] x$ (1)

• **Signal received BS $\rightarrow$ UAV :-** $r2 = \left[f \sqrt{L_{DIR}^{-1}} \right] x$ (2)

• Total energy received from direct path and indirect path is given by:-

$$E_{rxed} = \left| \sqrt{L_1} \sum_{i=1}^N |\alpha_i| |\beta_i| \right| + \left| |\gamma| \sqrt{L_2} \right|^2 E_s \quad (3)$$

where, $L_1 = L_{UAV}^{-1} L_{BS}^{-1}$ $L_2 = L_{DIR}^{-1}$ $\phi_i = \omega_i + \phi_i - \theta$
When CSI known to IRS

Frequency=700MHz

	Suburban	Urban	Dense Urban	Highrise Urban
(a_{LOS}, b_{LOS})	(11.53, 0.06)	(10.98, 0.05)	(9.64, 0.04)	(9.16, 0.03)
(a_{NLOS}, b_{NLOS})	(26.53, 0.03)	(23.31, 0.03)	(30.83, 0.04)	(32.13, 0.03)
c	0.77	0.63	0.37	0.06

• **Probability of LoS [2] :-** $P_{LOS_{UAV}}(\theta_{UAV}) = \frac{1}{1 + a_{UAV} e^{-b_{UAV}(\theta_{UAV} - a_{UAV})}}$ (4)

$P_{LOS_{BS}}(\theta_{BS}) = -a_{BS} e^{-b_{BS}\theta_{BS}} + c_{BS}$ (5)

$P_{LOS_{DIR}}(\theta_{DIR}) = -a_{DIR} e^{-b_{DIR}\theta_{DIR}} + c_{DIR}$ (6)

• **Path-loss (in decibels) [2] :-**

$L_{UAV}^{dB}(\theta_{UAV}, d_{UAV3D}) = (\eta_{LOS} - \eta_{NLOS}) P_{LOS_{UAV}}(\theta_{UAV}) + 20 \log \left(d_{UAV3D} \frac{4\pi}{\lambda} \right) + \eta_{NLOS}$ (7)

$L_{DIR}^{dB}(\theta_{DIR}, d_{DIR3D}) = (10K_{NLOS} \log(d_{DIR3D}) + \eta_{NLOS})(1 - P_{LOS_{DIR}}(\theta_{DIR})) + (10K_{LOS} \log(d_{DIR3D}) + \eta_{LOS})(P_{LOS_{DIR}}(\theta_{DIR})) - G_{sl} + 20 \log \left(\frac{4\pi}{\lambda} \right)$ (8)

$L_{BS}^{dB}(\theta_{BS}, d_{BS3D}) = (10K_{NLOS} \log(d_{BS3D}) + \eta_{NLOS})(1 - P_{LOS_{BS}}(\theta_{BS})) + (10K_{LOS} \log(d_{BS3D}) + \eta_{LOS})(P_{LOS_{BS}}(\theta_{BS})) - G_{sl} + 20 \log \left(\frac{4\pi}{\lambda} \right)$ (9)

$$X_1 = \sum_{i=1}^N |\alpha_i| |\beta_i| - \text{Gaussian} \quad \text{mean}, s = \frac{N\pi}{4} \quad \text{variance}, \sigma^2 = N(1 - \frac{\pi^2}{16})$$

$$Y_1 = |\gamma| - \text{Rayleigh distributed} \quad \lambda_D = \frac{1}{2}$$

$$CDF = \Pr(|\sqrt{K_1}X_1 + \sqrt{K_2}Y_1| < E_{thrs})$$

$$\text{Using : } F_{Z^2}(t) = F_Z(\sqrt{t})$$

$$f_{X_1}(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-s)^2}{2\sigma^2}}$$

$$f_{Y_1}(y) = \frac{y}{\lambda_D} e^{-\frac{y^2}{2\lambda_D}}$$

Suppose, $p = \sqrt{\frac{K_1}{K_2}}$ $K_1 = L_1 E_s$ $K_2 = L_2 E_s$ $a = \frac{1}{\sqrt{K_1}}$

$$c_1 = \sqrt{\sigma^2 p^2 + \lambda_D} \quad c_2 = \sqrt{2\sigma^2 \lambda_D} \quad c_3 = \sigma^2 a p \quad c_4 = \frac{\lambda_D a}{p} \quad c_5 = \lambda_D s$$

Outage Probability:- $P_{out} = \frac{1}{2} \left(\text{erf} \left(\frac{\frac{a}{p} \sqrt{E_{thrs}} - s}{\sqrt{2\sigma^2}} \right) + \text{erf} \left(\frac{s}{\sqrt{2\sigma^2}} \right) \right)$ (10)

$$-\frac{1}{2} \sqrt{\frac{\lambda_D}{c_1}} e^{-\frac{(a\sqrt{E_{thrs}} - \rho s)^2}{2c_1^2}} \text{erf} \left(\frac{c_4 \sqrt{E_{thrs}} - c_5}{c_1 c_2} \right) - \frac{1}{2} \sqrt{\frac{\lambda_D}{c_1}} e^{-\frac{(a\sqrt{E_{thrs}} - \rho s)^2}{2c_1^2}} \text{erf} \left(\frac{c_4 \sqrt{E_{thrs}} + c_5}{c_1 c_2} \right)$$

3. RESULTS AND DISCUSSIONS

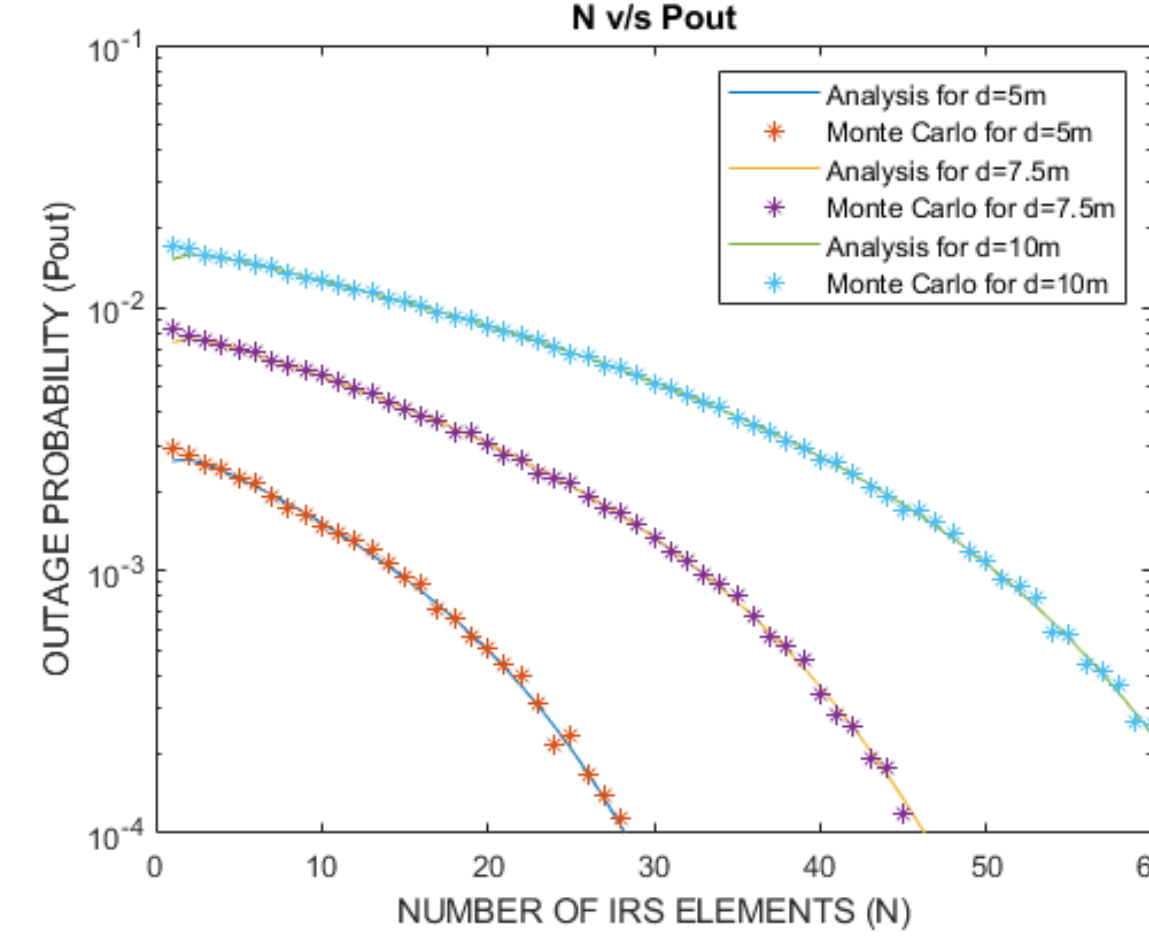


Figure 2: Plot between Number of IRS elements and Outage probability

❖ This plot is simulated and analyzed in urban environment parameters.

❖ From this plot, Outage probability (Pout) decreases with increase in number of IRS elements.

❖ As distance increases the outage probability also increases for same number of IRS elements.

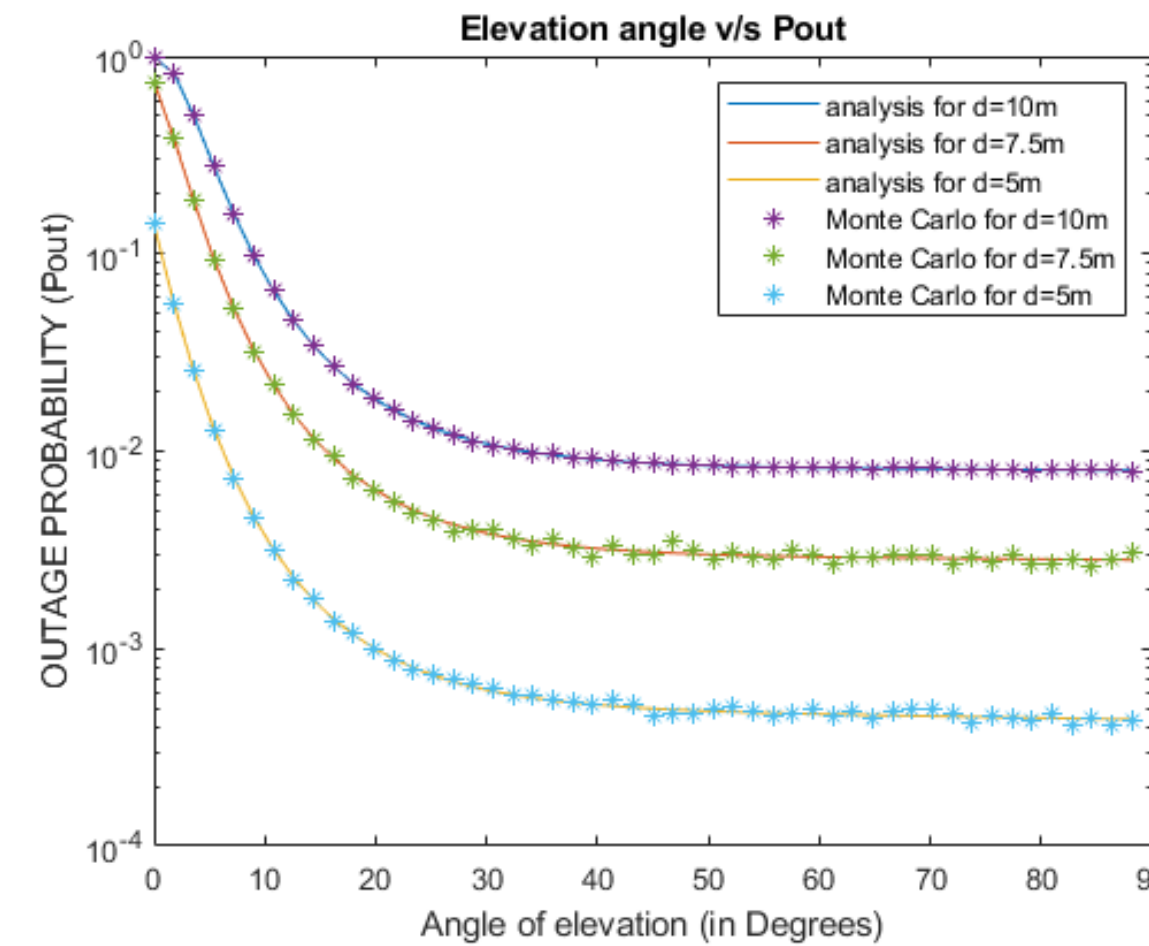


Figure 3: Plot between Elevation Angle and Outage Probability

❖ As elevation angle increases outage probability decreases as it increases LoS component from BS to UAV which is generally major contributor.

❖ For same elevation angle, outage probability increases with increase in distance between BS and UAV.

❖ No. of IRS elements, N= 20.

❖ The two plots above, proves that to use the system efficiently and to deliver maximum energy from BS to UAV, the UAV should be in LoS of BS i.e. maximum elevation angle and distance must be as less as possible so that the channel loss will be minimum.

4. FUTURE WORK

❖ The next study would be when IRS has no prior knowledge about CSI (Channel State Information) which suits practical systems and assumptions.

❖ To develop more practical scenario, discrete phase shifts would be studied which is more suitable for electronic systems.

5. REFERENCES

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